

Atlas HP-40

Hydraulic Press - Synthetic Service Manual

ASSET REFERENCE	SYN-HP-040
DOCUMENT STATUS	Revision 1.2 / Synthetic edition
DATA CLASS	Entirely fictional / non-production

1. Purpose and equipment summary

The fictional Atlas HP-40 is a single-station hydraulic forming press used in synthetic acceptance scenarios. Its pressure circuit includes a variable-displacement pump, proportional relief valve, return filter, oil cooler, and guarded ram area.

This manual exists only to exercise document upload, PDF extraction, page provenance, retrieval, and citation behavior in QIP. It does not describe real equipment and must not be used for physical maintenance.

Nominal operating envelope

Parameter	Synthetic value
Rated forming force	400 kN
Normal hydraulic pressure	155-165 bar
Reservoir volume	120 L
Recommended fluid	Synthetic ISO VG 46 test fluid
Normal oil temperature	35-58 C
Return-filter alarm	2.5 bar differential

2. Alarm and troubleshooting reference

Preserve observations before resetting an alarm. The table provides decision support for synthetic investigations; it does not establish root cause by itself.

H17	Oil temperature above 63 C for 90 seconds
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Recommended inspection: Check cooler airflow. If return-filter differential exceeds 2.5 bar, isolate the press and replace the filter element.

P08	Commanded pressure not reached
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Recommended inspection: Inspect for external leakage, confirm reservoir level, then verify relief-valve setting. Do not increase the setting above 165 bar.

R12	Ram retract time above 4.8 seconds
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Recommended inspection: Check return restriction and directional-valve response. A clogged return filter is the preferred first inspection.

INVESTIGATION EXERCISE	Scenario HP-A: After 37 minutes of cycling, oil temperature reaches 66 C and ram retract time rises to 5.4 seconds. The return-filter indicator reads 3.1 bar. Evidence supports inspecting and replacing the return filter before changing the relief-valve setting.
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3. Inspection and maintenance

Use the following synthetic intervals to test evidence capture, question answering, and citation to a specific page.

- At shift start: inspect guards, hoses, oil level, and emergency-stop function.
- Every 250 synthetic hours: record return-filter differential at 150 bar and clean the cooler intake screen.
- Every 1,000 synthetic hours: sample fluid, inspect hose date labels, and verify relief pressure with a calibrated gauge.
- After any H17 event: do not reset more than once without identifying the heat source.

Safety boundary

TEST DATA ONLY	Lockout, isolation, and competent-person requirements in this fictional document are illustrative. Never apply these values or procedures to real machinery.
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Suggested QIP test questions

- What observation should be preserved before changing a machine setting?
- Which threshold in the source supports the recommended first inspection?
- Is there enough evidence to state a confirmed root cause? Explain the uncertainty.